WakeSteer: Symmetry-Aware Deep Reinforcement Learning for Wind-Farm Wake Steering

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AI Course Mini-Project, Assignment 2 (Real-Life Use Cases of RL)
Code, trained agents & reproduction: https://github.com/Rishi-0312/AI-Assignment-2

***Abstract*—Wakes of upstream turbines cut the output of downstream machines by up to 75% in the farm we study. *Wake steering* yaws upstream rotors away from the wind to deflect their wakes, but the industrial look-up table (LUT) of set-points, optimised offline with an engineering model, ignores sensor noise, yaw-drive limits and actuator wear. We cast farm-level yaw control as a Markov decision process with stochastic wind, noisy sensing, rate-limited yaw drives and a wear penalty, simulated with NREL's FLORIS, and solve it with PPO (implemented from scratch) using an exactly rotation-equivariant policy. On 512 held-out one-hour episodes of a 3×3 farm the agent raises energy yield by 2.92% over greedy control, versus 2.53% for a LUT optimised with the very wake model it is tested in and 2.72% for that LUT with tuned measurement filtering, while yawing 49% less than the LUT.**

I. Introduction

Large wind farms lose 10–20% of their energy to wakes [2]. Yawing a turbine by γ costs it roughly a $\cos^{1.88}\gamma$ share of its power but bends its wake off downstream rotors, which can recover far more [3], [6], [7]. Choosing all offsets is a *sequential* problem: the wind drifts and is measured noisily, yaw drives are slow and travel wears them. Offline LUTs [5] treat each instant in isolation and trust perfect models and measurements; RL learns closed-loop behaviour from interaction [11]–[13]. *Contributions:* (i) an MDP of dynamic wake steering with realistic sensing and actuation; (ii) a batched, Gymnasium-compatible FLORIS environment fast enough for on-policy RL on a laptop; (iii) PPO with an exactly rotation-equivariant policy; (iv) a paired, held-out evaluation against greedy and model-based control with ablations.

II. MDP Formulation

Simulator. Nine NREL 5-MW turbines [8] ($D = 126\,\mathrm{m}$) on a 3×3 grid with $5D$ spacing, turbulence intensity (TI) 6%; powers from FLORIS v4 [1], Gauss–Curl-Hybrid model (Gaussian deficit, deflection, secondary steering, yaw-added recovery [4]). Step $\Delta t = 60\,\mathrm{s}$; episode $T = 60$ steps (one hour).

Exogenous wind. Direction: persistent random walk $\theta_{t+1} = \theta_t + \delta_{t+1}$, $\delta_{t+1} = 0.7\,\delta_t + 0.8^\circ\epsilon_t$, $\theta_0 \sim \mathcal{U}[0^\circ, 360^\circ)$. Speed: Ornstein–Uhlenbeck $U_{t+1} = U_t + 0.1(\bar{U} - U_t) + 0.25\,\epsilon'_t$, $\bar{U} \sim \mathcal{U}[5, 12]$ m/s. Controllers only see $\hat{\theta}_t = \theta_t + \mathcal{N}(0, 2^{\circ\,2})$ and $\hat{U}_t = U_t + \mathcal{N}(0, 0.2^2)$.

State space $\mathcal{S}$. The Markov state holds the true wind $(\theta, \delta, U, \bar{U})$, the measurements and the nacelle headings h_i. The agent observes $o_t \in \mathbb{R}^{22}$ (a POMDP):

$$o_t = \Big[\sin\hat{\theta}_t,\ \cos\hat{\theta}_t,\ \tfrac{\hat{U}_t - 8.5}{3.5},\ \tfrac{\Delta\hat{\theta}_t}{5^\circ},\ \big\{\tfrac{\hat{\gamma}_{i,t}}{25^\circ}\big\}_{i=1}^{9},\ \big\{\tfrac{P_{i,t}}{P_{\mathrm{rated}}}\big\}_{i=1}^{9}\Big],$$

with measured yaw offsets $\hat{\gamma}_i = \mathrm{wrap}(\hat{\theta} - h_i)$ and turbine outputs P_i, which depend on the *true* wind and thus act as a noise-free wake sensor.

Action space $\mathcal{A}$. Continuous $a_t \in [-1, 1]^9$, yaw-offset set-points $\gamma_i^\star = 25^\circ a_i$. Each yaw drive turns at most $\omega = 0.3^\circ$/s: $h_{i,t+1} = h_{i,t} + \Delta h_{i,t}$ with $\Delta h_{i,t} = \mathrm{clip}\big(\mathrm{wrap}(\hat{\theta}_t - \gamma_i^\star - h_{i,t}), \pm\omega\Delta t\big)$. Then the wind evolves and FLORIS returns $P_{i,t+1}$ for the true misalignments $\gamma_i = \theta_{t+1} - h_{i,t+1}$.

Reward $\mathcal{R}$ (in % of farm capacity $P_{\mathrm{cap}} = 45\,\mathrm{MW}$):

$$r_t = \tfrac{100}{P_{\mathrm{cap}}}\Big(\textstyle\sum_i P_{i,t+1} - P^{\mathrm{greedy}}(\theta_{t+1}, U_{t+1})\Big) - c\textstyle\sum_i |\Delta h_{i,t}|. \tag{1}$$

P^{greedy} is the all-aligned farm's power in the same wind (tabulated). No controller can affect the wind, so it shifts every policy's return equally: the optimal policy is unchanged while the variance due to the U^3 power law vanishes. The term with $c = 0.005$/deg prices yaw-drive wear and discourages chattering.

III. Environment, Algorithm and Baselines

Environment. A FLORIS call has ≈20 ms overhead, so 256 environments advance in lock-step with *one* vectorised FLORIS call per step, split over worker processes (3.8k steps/s on an i7-12700H laptop); a Gymnasium interface (`WakeSteer-v0`) is included.

Symmetry-aware policy. The grid is invariant under 90° rotation and FLORIS works in the wind frame, so $P_i(\theta, \boldsymbol{\gamma}) = P_{\pi(i)}(\theta - 90^\circ, \Pi\boldsymbol{\gamma})$ for a turbine permutation π (verified to 10^{-6}). Each observation is rotated by k quarter turns into the sector $\hat{\theta} - 90^\circ k \in [15^\circ, 105^\circ)$ (cut where steering gains vanish), turbine features are permuted by π^k and actions permuted back: the policy is equivariant by construction and learns only a quarter of the wind rose.

PPO [9]. Gaussian policy, separate 2×256 tanh actor and critic, Fourier features $\sin k\theta, \cos k\theta$ ($k \in \{1, 2, 3, 4, 8\}$). Clipped surrogate ($\epsilon = 0.2$), GAE [10] ($\lambda = 0.9$), 256 envs×30 steps per update, 10 epochs×8 minibatches, Adam 3×10^{-4} (annealed), KL early stop at 0.02, per-network gradient clipping, bootstrapping at time-limit truncations, random initial offsets; 3×10^6 steps per run (≈1 h), keeping the checkpoint with the

TABLE I: Test on 512 unseen one-hour episodes (identical weather). Training: PPO 3×3M steps, $\gamma = 0.95$ 3×3.6M, plain MLP 5M, others 3M.

Controller	Energy gain vs greedy [%]	Yaw travel [°/turb./h]	Side switches/h	Reward per step
Greedy (baseline)	+0.00	143	0.4	−0.15
LUT (FLORIS-SR)	+2.53	290	2.2	+0.87
LUT + filter (α=0.6)	+2.72	185	1.4	+1.04
PPO (ours, 3 seeds)	**+2.92±0.005**	**147±2**	**1.3**	**+1.15**
– long horizon ($\gamma = 0.95$)	+2.52	169	1.5	+0.96
– plain MLP ($\gamma = 0.95$)	+2.25	169	1.7	+0.84
– no power inputs	+2.74	159	1.4	+1.07
– no wear penalty	+2.92	173	1.4	+1.14

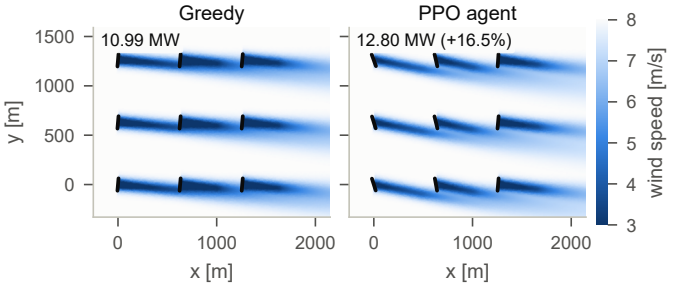


Fig. 1: Hub-height flow, wind from 275° at 8 m/s: greedy control vs. settled PPO set-points.

best return on fixed validation weather. The discount γ is a training hyper-parameter (the objective is undiscounted energy) selected from $\{0, 0.5, 0.8, 0.95\}$ on validation weather. PPO fits: actions are continuous and 9-D (a discretised DQN would face $\gg 10^8$ joint actions).

Baselines. *Greedy*: zero offsets (industry standard). *LUT*: offsets optimised offline by FLORIS' Serial-Refine method [5] on a $1° \times 1$ m/s grid (3 960 optimisations) with the *same* wake model as the test, applied at the measured wind—perfect model knowledge. *LUT+filter*: that LUT fed with exponentially smoothed measurements, the nacelles tracking the smoothed direction ($\alpha = 0.6$, tuned on validation weather). All share the yaw dynamics and face *identical* wind in 512 one-hour test episodes of unseen weather (paired comparison).

IV. Results

Main result. PPO ($\gamma = 0.5$) gains 2.92±0.005% energy over greedy control (std over 3 seeds; random set-points: −2.68%), against 2.53% for the LUT and 2.72% for LUT+filter (Table I, Fig. 1). Paired over identical weather it beats LUT+filter by 0.19±0.017 percentage points (s.e.; better in 72% of episodes) and the LUT by 0.39±0.026 (84%), yawing 147° per turbine-hour instead of 290° and 185°. The LUT chases noisy measurements near alignments; the agent holds its deflection side: as the wind veers slowly through the 270° row alignment it switches at 272.7° veering right but 267.4° veering left, a 5.3° hysteresis band (LUT: 1.8°, yaw lag only).

Learning and ablations. *Discount:* best validation rewards were 1.07, 1.08, 1.01 and 0.88 for $\gamma = 0$, 0.5, 0.8, 0.95 (test gains 2.89, 2.92, 2.79, 2.52%; Fig. 2b). Long-horizon returns are dominated by the exogenous wind while a yaw drive reaches any set-point in 1–3 steps, so a two-step horizon captures yaw

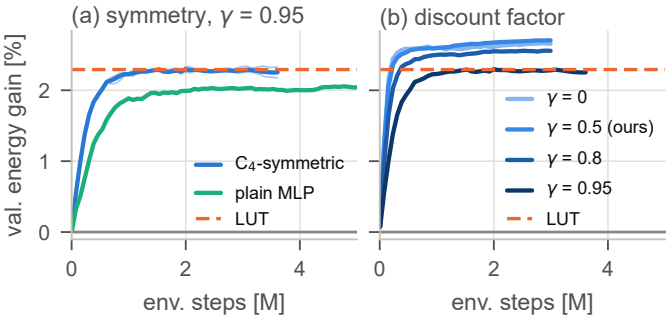


Fig. 2: Validation energy gain during training (thin: individual seeds).

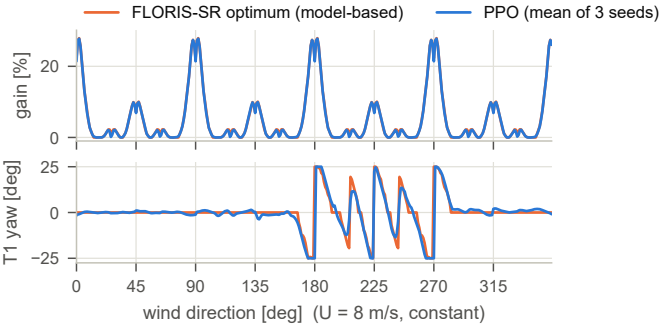


Fig. 3: Steady, noise-free wind at 8 m/s: the agent recovers 97% of the model-optimal gain (5.27% vs 5.40%) without ever being shown it, including the sign switches of the optimal yaw of corner turbine T1 (bottom).

manoeuvres at low variance. *Symmetry* (at $\gamma = 0.95$, Fig. 2a): about 3× faster learning and a higher plateau than a plain MLP (test 2.52% vs 2.25%). *Inputs and reward:* without turbine-power inputs the gain drops to 2.74%; without the wear penalty it is unchanged (2.92%) but yaw travel rises to 173°/h.

Robustness (Fig. 4). With perfect direction measurements the raw LUT is best (3.39% vs 3.05%), but it collapses with noise (0.38% at $\sigma = 6°$) while PPO keeps 1.83% (LUT+filter 1.40%). Under the cumulative-curl wake model (3.05, 3.22, 3.41% for LUT, LUT+filter, PPO) and at TI = 10% (1.18, 1.25, 1.44%) the ranking is unchanged.

V. Conclusion

PPO beats greedy control and model-optimised LUTs under realistic sensing and actuation, with less yaw travel and 0.10 ms decisions (online optimisation: 888 ms). Limitations: steady-state wakes, one layout.

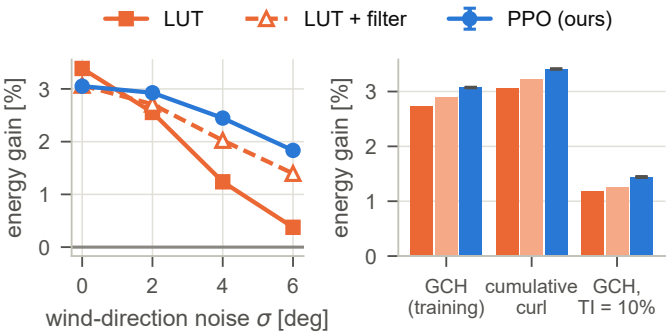


Fig. 4: Robustness without retraining: wind-direction noise (left); wake-model and TI mismatch (right).

REFERENCES

[1] NREL, "FLORIS: FLOw Redirection and Induction in Steady State, v4," 2024. https://github.com/NREL/floris

[2] R. J. Barthelmie *et al.*, "Modelling and measuring flow and wind turbine wake losses in large offshore wind farms," *Wind Energy*, vol. 12, no. 5, pp. 431–444, 2009.

[3] M. Bastankhah and F. Porté-Agel, "Experimental and theoretical study of wind turbine wakes in yawed conditions," *J. Fluid Mech.*, vol. 806, pp. 506–541, 2016.

[4] J. King *et al.*, "Control-oriented model for secondary effects of wake steering," *Wind Energy Science*, vol. 6, pp. 701–714, 2021.

[5] P. Fleming *et al.*, "Serial-refine method for fast wake-steering yaw optimization," *J. Phys.: Conf. Ser.*, vol. 2265, 032109, 2022.

[6] M. F. Howland, S. K. Lele and J. O. Dabiri, "Wind farm power optimization through wake steering," *PNAS*, vol. 116, no. 29, pp. 14495–14500, 2019.

[7] P. Fleming *et al.*, "Initial results from a field campaign of wake steering applied at a commercial wind farm – Part 1," *Wind Energy Science*, vol. 4, pp. 273–285, 2019.

[8] J. Jonkman, S. Butterfield, W. Musial and G. Scott, "Definition of a 5-MW reference wind turbine for offshore system development," NREL/TP-500-38060, 2009.

[9] J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[10] J. Schulman, P. Moritz, S. Levine, M. Jordan and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Proc. ICLR*, 2016.

[11] P. Stanfel, K. Johnson, C. J. Bay and J. King, "A distributed reinforcement learning yaw control approach for wind farm energy capture maximization," in *Proc. American Control Conf.*, 2020, pp. 4065–4070.

[12] H. Dong, J. Zhang and X. Zhao, "Intelligent wind farm control via deep reinforcement learning and high-fidelity simulations," *Applied Energy*, vol. 292, 116928, 2021.

[13] E. Kadoche, S. Gourvénec, M. Pallud and T. Levent, "MARLYC: Multi-agent reinforcement learning yaw control," *Renewable Energy*, vol. 217, 119129, 2023.